

Control work, and the quarter review

The axioms, the positions, the polyhedra, the sections and the trigonometry — all in one paper, then the map that ties them together.

LESSON 17–18

2 × 40 MIN

GEOMETRY 10, NAZORAT ISHI 1

IGCSE E4 REVIEW

LESSON CLOCK

3

42'

12'

20'

10'

3

Instructions	3 min
The paper	42 min
Answers and self-marking	12 min
Rewrite and classify	20 min
Concept map	10 min
Homework	3 min

LEARNING OBJECTIVES

- Apply the whole quarter in one assessment.
- Present a construction with the rule quoted at each step.
- Classify each lost mark and rewrite the solution.
- Build a concept map of solid geometry so far.

TEXTBOOK

National	pp. 79–82
Cambridge	Core & Extended, pp. 313–316
Workbook	Control paper G1

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	State the three axioms of space and say what each forbids	3	L4–5
2	In a cube, list all edges skew to CD	3	L6–7
3	A polyhedron has $F = 12$ and $E = 30$. Find V , and name it if regular	3	L8–9
4	Construct the section of a cube through A , C and B_1 ; name the shape and find its area for side 6	6	L12–13
5	Cuboid $6 \times 8 \times 10$: space diagonal and its angle with the base	5	L14–16
6	Square pyramid base 10, height 12: the angle between a lateral face and the base	5	L14–16
7	Count the planes of symmetry of a cube and of a $3 \times 3 \times 7$ cuboid	5	L3

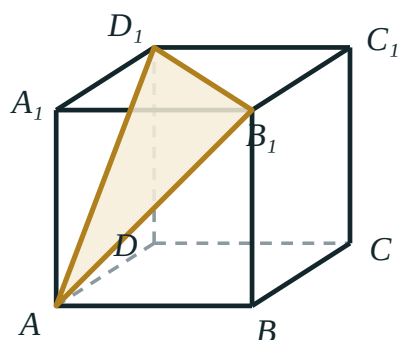
CONSTRUCTIONS ARE MARKED ON REASONS

Q4 gives 2 marks for the correct figure and 4 for the justification. A section drawn without R1/R2/R3 quoted scores a third of the question.

The concept map

Six boxes, and the links between them written as sentences:

- **axioms** → **positions of lines and planes** — “S3 is why two planes meet in a line”
- **positions** → **skew lines** — “the case the plane does not have”
- **polyhedra** → **Euler** — “ $F + V - E = 2$ for every convex solid”
- **polyhedra** → **nets** → **surface area**
- **sections** → **R1, R2, R3** — “parallel faces give parallel cuts”
- **3-D trigonometry** → **extract a triangle** — “every 3-D question is a plane question”



the section
A B₁ D₁ is
an equilateral
triangle

*One cube carries four of the six boxes: positions,
sections, symmetry and 3-D trigonometry.*

Looking forward

Quarter II is parallelism of lines and planes — every proof will quote the three axioms of Lesson 4–5 by name, and every construction will use the section rules of Lesson 12–13. Neither set is ever re-taught; they are simply used.

ONE HABIT TO CARRY FORWARD

Draw the solid large, mark the right angles first, then extract the triangle. Almost every mark lost in Q5 and Q6 was lost by working from a diagram three centimetres wide.

02 Worked examples

EXAMPLE 1 Model answer, Q4: section of a cube through A , C , B_1 , side 6.

- ① AC in the base face
R1

- ② AB_1 in face ABB_1A_1 , CB_1 in face BCC_1B_1
R1 twice

- ③ All three are face diagonals, $6\sqrt{2}$
Equilateral triangle

- ④ $S = \frac{\sqrt{3}}{4} \times 72 = 18\sqrt{3}$

ANSWER

Equilateral triangle, area $18\sqrt{3} \approx 31.2$

EXAMPLE 2 Model answer, Q6: square pyramid base 10, height 12.

- ① Apothem of the base 5.

- ② Slant height $\sqrt{144 + 25} = 13$.

- ③ $\cos \theta = \frac{5}{13}$
Both lines $\perp$ to the base edge.

- ④ $\theta \approx 67.4^\circ$

ANSWER

$\approx 67.4^\circ$

EXAMPLE 3 Model answer, Q3: $F = 12$, $E = 30$.

① $V = 2 - F + E = 2 - 12 + 30$

② $V = 20$

③ 12 faces, 20 vertices, 30 edges.
The dodecahedron.

ANSWER

$V = 20$ — the regular dodecahedron

03 Interactive model

Work Q4 and Q6 on the board before the rewrite hour begins.

The quarter in ten questions

Three non-collinear points determine:

no plane

one plane

two

infinitely many

Edges of a cube skew to a given edge:

2

3

4

6

$F = 12, E = 30$ gives V :

12

16

20

24

Section through A , C , B_1 of a cube:

square

rectangle

equilateral triangle

hexagon

Space diagonal of a $6 \times 8 \times 10$ cuboid:

$10\sqrt{2}$

24

14

20

Planes of symmetry of a cube:

3

6

9

12

Planes of symmetry of a $3 \times 3 \times 7$ cuboid:

3

4

5

9

A plane cuts two parallel faces in:

equal lines

parallel lines

perpendicular lines

points

The maximum sides of a cube section:

4

5

6

8

The angle between a line and a plane uses:

the normal

the projection

any line in the plane

the edge

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Assessment	Baholash	Оценивание
Justification	Asoslash	Обоснование
Construction step	Yasash bosqichi	Шаг построения
Concept map	Tushunchalar xaritasi	Карта понятий
Revision	Takrorlash	Повторение
Target	Maqsad	Цель
Self-assessment	O'z-o'zini baholash	Самооценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A construction scores most of its marks for:

neatness

the justification

colour

speed

Euler's formula fails for:

a cube

a prism

a solid with a hole

a pyramid

Before any 3-D trigonometry you should:

guess

extract and redraw the triangle

measure

use a calculator

Skew lines differ from parallel lines because:

they meet

no plane contains both

they are equal

they are perpendicular

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. State axiom S1

ANSWER Three non-collinear points determine exactly one plane

2. Edges skew to CD in a cube

ANSWER $AA_1, BB_1, A_1D_1, B_1C_1$

3. $F = 12, E = 30$; find V

ANSWER 20

4. Space diagonal of a $6 \times 8 \times 10$ cuboid

ANSWER $10\sqrt{2}$

5. Planes of symmetry of a cube

ANSWER 9

6. Section through A, C, B_1

ANSWER equilateral triangle

7. Slant height of a pyramid, base 10, height 12

ANSWER 13

Medium · the standard question

7 PROBLEMS

1. Area of the section through A , C , B_1 for side 6

ANSWER $18\sqrt{3}$

2. Angle of a $6 \times 8 \times 10$ space diagonal with the base

ANSWER 45°

3. Angle between a lateral face and the base, pyramid 10/12

ANSWER $\approx 67.4^\circ$

4. Planes of symmetry of a $3 \times 3 \times 7$ cuboid

ANSWER 5

5. F , V , E of an octagonal prism

ANSWER 10, 16, 24

6. Diagonal section area, cube side 6

ANSWER $36\sqrt{2}$

7. Angle of a lateral edge with the base, pyramid base 10 height 12

ANSWER $\approx 59.5^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Cube side 6: area of the midpoint hexagonal section

ANSWER $27\sqrt{3} \approx 46.8$

2. Prove no cube section is a heptagon

ANSWER Six faces give at most six sides

3. A solid has 32 faces, 12 pentagons and 20 hexagons, 3 at each vertex. Find V, E

ANSWER $V = 60, E = 90$

4. Cuboid $a \times a \times b$: find the angle between the space diagonal and a base edge

ANSWER $\arccos \frac{a}{\sqrt{2a^2 + b^2}}$

5. Regular tetrahedron edge a : find the angle between two faces

ANSWER $\approx 70.53^\circ$

6. Construct the section of a cube through the midpoints of AB, BC and CC_1

ANSWER A pentagon

7. Explain why a plane cannot meet a face of a convex solid in two separate segments

ANSWER Convexity makes the intersection connected

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter II.

1. Rewrite in full every control-work question that lost a mark, with reasons.
2. Finish the concept map with all six links written as sentences.
3. Cuboid $5 \times 12 \times 9$: find the space diagonal, its angle with the base and its angle with the longest edge.

4. Write your target for Quarter II in one checkable sentence.